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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Vellore – 632014, Tamil Nadu, India

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Win Semester 2025-2026

Continuous Assessment-1

Start of Patent Creation

SLOT : D1+TD1
Programme Name : B.Tech.
Course Name & code : Introduction to Internet of Things & CSE1030
Class Number : VL2025260505493
Faculty Name : Dr. Mehfooza M
Team No : 99

Page 1: Invention Overview & Inventor Information

Part A: Inventor & Project Details

Field	Details
Project Title	External IoT-Based Predictive Thermal Stress Monitoring System for EV Batteries
Invention Domain	Internet of Things (IoT), Embedded Systems, Electric Vehicles, Battery Safety
Academic Year	2025–2026
Student Name(s)	Shakeer Ahmed K Kathireswar V M Gunal Giridharan M G
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School/Center	School of Advanced Science (SAS)

Part B: Problem Statement & Motivation

Problem Statement

Electric Vehicles (EVs) are increasing rapidly, especially low-cost EVs such as e-scooters and e-rickshaws. Battery overheating is one of the major causes of EV fires. Existing safety systems mainly depend on internal Battery Management Systems (BMS), which are not accessible in many low-cost or old EVs.

Most systems trigger alerts only when a fixed temperature limit is crossed. This does not provide early warning before a dangerous situation occurs..

- **Current Challenges**

- Internal BMS data is not accessible in many EVs.
- Fixed temperature limit alerts are late.
- Advanced systems are costly.
- Low-cost EVs lack predictive safety features.

- **Target Users / Environment**

- Electric vehicle drivers.
- Delivery and fleet EVs.
- Electric scooters and e-rickshaws.
- Hot climate areas like India.

- **Motivation**

Battery fire incidents are dangerous and can cause accidents and loss of life. Providing an early warning system can help drivers stop the vehicle before a serious problem occurs. This project aims to improve EV safety in a simple and affordable way.

Part C: Objectives of the Invention

1. To monitor EV battery surface temperature using an external sensors.
2. To predict battery risk based on temperature rise behaviour.
3. To calculate a Thermal Stress Index to estimate risk level.
4. To alert the driver using LED indicators and buzzer.
5. To design a low-cost and easy-to-use safety system for EVs.

Part D: Novelty & Uniqueness

Key Differentiators

1) Rate-of-Temperature-Change Based Prediction:

The system predicts battery risk by analysing how fast the battery temperature rises instead of using a fixed temperature limit.

2) Thermal Stress Index (TSI):

A Thermal Stress Index is calculated to convert temperature rise behavior into low, medium, or high risk levels.

3) External and Plug-and-Play Design:

The system works independently of the internal BMS and can be externally attached to existing EV batteries.

4) Offline-First Safety Operation:

All safety decisions are made locally on the controller, while internet connectivity is used only for monitoring.

5) Low-Cost and Driver-Friendly Alerts:

The system communicates battery risk to the driver using simple visual (LEDs) and audio (buzzer) alerts, making it easy to understand and suitable for low-cost EV applications..

Innovation Statement

This invention presents a low-cost, external, and predictive EV battery safety system that estimates future risk using temperature rise behavior. The system enhances EV safety without accessing internal BMS data and provides early, driver-friendly warnings.